

Teaching Guide: Building a Monthly DiD Dataset from USITC Data

Honors Thesis Workflow

February 17, 2026

1 Goal

Build an analysis-ready monthly panel for Difference-in-Differences (DiD), where treatment is based on pre-policy NTR-gap exposure and policy timing is January 2002.

2 Required Datasets

File	Purpose
raw_data/China_Imports_1996_2005.csv	Main trade file from USITC DataWeb with monthly columns (January to December), product codes, values, and quantities.
transformed_data/data_files_aer_2013/tar_val.dta panel_hts10_monthly.csv	Tariff/NTR-gap source. In this project, key variables include hs8 , year , spread , and v . Intermediate monthly panel at HTS10 level (created during cleaning).
did_panel_monthly.csv	Final DiD-ready panel (you will create this).

3 General Tasks

1. Build a clean monthly panel from raw trade data.

- Keep product code as string (preserve leading zeros).
- Reshape month columns to long format.
- Construct monthly **date** (YYYY-MM-01).

2. Keep unit logic consistent.

- Do not mix quantities across different unit descriptions.
- Use **quantity_description** from first-unit quantity records.
- Drop known HTS10 codes with unit-description switches.

3. Aggregate to analysis level (HS8 monthly).

- Build (*hs8*, *date*) panel.
- Sum **customs_value**.

- Sum quantity only when unit is consistent within $(hs8, date)$.
4. **Construct treatment exposure from tariffs.**
 - Read `tar_val.dta`.
 - Use year 1999 `spread` as pre-period exposure.
 - Collapse to HS8 (weighted by `v` if desired) and merge to trade panel.
 5. **Create DiD variables.**
 - `post` = 1 if `date` $\geq$ 2002-01-01, else 0.
 - `highgap` = 1 if `gap_pre` above/equal median.
 - `treated` = `highgap` $\times$ `post`.
 - `event_time`: months relative to 2002-01-01.
 6. **Estimate models and create outputs.**
 - Run DiD ladder with clustered SE at product level.
 - Run event studies.
 - Export tables and figures.

4 Core DiD Specification

For product p in month t :

$$y_{pt} = \alpha_p + \gamma_t + \beta(\text{highgap}_p \times \text{post}_t) + \varepsilon_{pt}.$$

Where y_{pt} is one of:

- `ln_customs_value`
- `ln_first_unit_qty`
- `ln_unit_value`

5 Pseudocode (Data Construction)

INPUT:

```
trade_file = raw_data/China_Imports_1996_2005.csv
tariff_file = transformed_data/data_files_aer_2013/tar_val.dta
```

STEP A: Load trade panel

```
read trade data
keep China rows
ensure hts10 is 10-digit string
parse date
```

STEP B: Unit quality control

```
drop known switching hts10 list
```

```
create hs8 = first 8 digits of hts10
```

STEP C: Aggregate to hs8-month

```
group by (hs8, date)
  customs_value = sum(customs_value)
  unique_unit_count = number of distinct quantity_description
  unit_consistent = unique_unit_count <= 1
  first_unit_qty_hs8 = sum(first_unit_qty) only if unit_consistent
  unit_value = customs_value / first_unit_qty_hs8 if qty > 0
```

STEP D: Build gap_pre from tariff data

```
read tariff_file
filter year == 1999
map code to hs8
gap_pre_hs8 = weighted mean(spread, weights = v) by hs8
```

STEP E: Merge and create DiD indicators

```
merge gap_pre_hs8 onto hs8-month panel
post = 1(date >= 2002-01-01)
highgap = 1(gap_pre >= median(gap_pre))
treated = post * highgap
event_time = month_distance(date, "2002-01-01")
```

STEP F: Logs

```
ln_customs_value = log1p(customs_value)
ln_first_unit_qty = log1p(first_unit_qty_hs8) when not missing
ln_unit_value = log(unit_value) when unit_value > 0
```

OUTPUT:

```
did_panel_monthly.csv
```

6 Pseudocode (Estimation)

Load did_panel_monthly.csv

product = hs8

Filter: non-missing outcome, post, highgap, date

For each outcome in [ln_customs_value, ln_first_unit_qty, ln_unit_value]:

```
m1: outcome ~ highgap * post
m2: outcome ~ highgap * post | product
m3: outcome ~ highgap * post | date
m4: outcome ~ highgap * post | product + date
cluster SE by product
save ladder table
```

Event study:

```
outcome ~ i(event_time, highgap, ref = -1) | product + date
```

```
cluster SE by product
save event-study plot with x-axis labeled in calendar years
```

7 Minimum Diagnostics Checklist

- No duplicate keys at $(hs8, date)$.
- Date range is 1996-01 to 2005-12.
- Report share of `unit_consistent == 0`.
- Report number of unique HS8 codes.
- Summarize `gap_pre`, `customs_value`, and `unit_value`.

8 Practical Notes

- Always preserve leading zeros in HS codes.
- Never sum quantity across mixed units.
- Keep every intermediate output; reproducibility matters.
- If results differ across machines, verify package versions and file paths first.